

AArch64 Assembly Programming: Arithmetic, Logic, & Control Flow

ALU mechanics, shifted-register operands, bitwise operations,
condition flags, structured branches, and conditional selection

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Computer Systems Architecture — AArch64 Reference Platform

Topic 4: Core Lecture Outline

① ALU Operations: Arithmetic, Shifting, & Bitwise Logic

- Addition, subtraction, multiplication, division, and shifted-register operands.
- Bitwise logic (and, orr, eor, bic) and bitfield extraction (ubfx).

② Condition Flags & Comparison Mechanics

- NZCV flag generation, comparison instructions (cmp, tst), and signed vs. unsigned branches.

③ Control Flow: Branches & Branchless Selection

- Translating loops and if-else structures; branch-and-test (cbz/cbnz).
- Branch penalties and branchless conditional selection (csel, cset).

PART 1

ALU Operations: Arithmetic, Shifting, & Bitwise Logic

Arithmetic Primitives: Addition & Subtraction

- Many AArch64 data-processing instructions use a **3-operand non-destructive format**. Flag-setting forms such as adds and subs update NZCV; add and sub do not.

Instruction	Operation	Flags Updated?
add x0, x1, x2	$x_0 \leftarrow x_1 + x_2$	No
adds x0, x1, x2	$x_0 \leftarrow x_1 + x_2$	Yes (NZCV)
sub x0, x1, x2	$x_0 \leftarrow x_1 - x_2$	No
subs x0, x1, x2	$x_0 \leftarrow x_1 - x_2$	Yes (NZCV)
neg x0, x1	$x_0 \leftarrow 0 - x_1$	No; alias of sub x0, xzr, x1

Immediate Constant Arithmetic

add x0, x1, #42 uses a 12-bit immediate. The add/sub immediate form can also shift that field left by 12 bits. The corresponding w-register form performs 32-bit arithmetic.

Shifted-Register Operands and Shift Operations

- Many AArch64 data-processing instructions can encode a shifted register operand directly in the instruction.
- Execution latency for shifted operands is implementation-dependent.

Shift Type	Mnemonic	Operation Meaning
Logical Shift Left	lsl #k	Multiply by 2^k modulo the register width; zeros enter the low bits.
Logical Shift Right	lsr #k	Unsigned floor division by 2^k ; zeros enter the high bits.
Arithmetic Shift Right	asr #k	Sign-preserving right shift; the sign bit is replicated into the high bits.

Integrated Shift-Add Examples

```
add x0, x1, x2, lsl #3     $x_0 \leftarrow x_1 + (x_2 \times 8)$ 
sub x0, x1, x2, lsr #1     $x_0 \leftarrow x_1 - \lfloor x_2/2 \rfloor$  for unsigned  $x_2$ 
```

Integer Multiplication & Division

- AArch64 defines instructions for integer multiplication and division, leaving the underlying microarchitectural execution units implementation-dependent.

Instruction	RTL Semantic Action	Description
<code>mul x0, x1, x2</code>	$x_0 \leftarrow x_1 \times x_2$	Lower 64 bits of the product
<code>madd x0, x1, x2, x3</code>	$x_0 \leftarrow x_3 + (x_1 \times x_2)$	Multiply-Add
<code>msub x0, x1, x2, x3</code>	$x_0 \leftarrow x_3 - (x_1 \times x_2)$	Multiply-Subtract
<code>sdiv x0, x1, x2</code>	$x_0 \leftarrow x_1 / x_2$	Signed division
<code>udiv x0, x1, x2</code>	$x_0 \leftarrow x_1 / x_2$	Unsigned division

Division by Zero

AArch64 integer division by zero does **not** raise an exception; it returns 0 in the destination register. For signed division, the overflow case $\text{INT64_MIN}/(-1)$ also does not trap; the destination receives the architecturally defined 64-bit result.

Bitwise Logic Primitives

- Bitwise operations apply independently to each corresponding bit position.

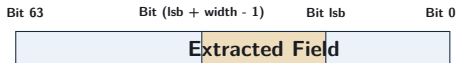
Instruction	Operation	Bitwise RTL Formula
<code>and x0, x1, x2</code>	Bitwise AND	$x_0 \leftarrow x_1 \wedge x_2$
<code>orr x0, x1, x2</code>	Bitwise OR	$x_0 \leftarrow x_1 \vee x_2$
<code>eor x0, x1, x2</code>	Bitwise XOR	$x_0 \leftarrow x_1 \oplus x_2$
<code>bic x0, x1, x2</code>	Bit Clear (AND NOT)	$x_0 \leftarrow x_1 \wedge (\sim x_2)$
<code>mvn x0, x1</code>	Bitwise NOT	$x_0 \leftarrow \sim x_1$ (Alias: <code>orn x0, xzr, x1</code>)

Bit Clear (bic) Idiom

`bic x0, x1, x2` sets bits to 0 in x_1 wherever the mask in x_2 has 1s.

Bitfield Extraction: `ubfx`

- AArch64 provides bitfield instructions that extract packed fields without explicit mask-and-shift sequences.
- `ubfx xd, xn, #lsb, #width`: unsigned bitfield extract.



Example: Extract 8-bit Field Starting at Bit 16

Mask + shift: `lsr x0, x1, #16 → and x0, x0, #0xff`

Bitfield extract: `ubfx x0, x1, #16, #8`

PART 2

Condition Flags & Comparison Mechanics

The NZCV Condition Flags

- Flag bits reside in pstate and record status of the most recent instruction that updates NZCV.
- Common flag-setting forms include instructions with an “s” suffix and explicit comparisons.

Flag	Name	Condition Set to 1
N	Negative	The most-significant bit of the result is 1.
Z	Zero	Result is exactly zero.
C	Carry	Unsigned carry-out (addition) or no borrow (subtraction).
V	Overflow	The signed mathematical result cannot be represented in the destination width.

Carry Flag in Subtraction

In AArch64 subtraction ($A - B$), $C = 1$ indicates $A \geq B$ (no borrow required). $C = 0$ indicates $A < B$ (unsigned borrow occurred).

Comparison Instructions: `cmp` and `tst`

- Comparison aliases set NZCV flags and discard the computed result.

Instruction	Equivalent Operation	Flags Set
<code>cmp x0, x1</code>	<code>subs xzr, x0, x1</code>	NZCV from $x_0 - x_1$
<code>cmn x0, x1</code>	<code>adds xzr, x0, x1</code>	NZCV from $x_0 + x_1$
<code>tst x0, x1</code>	<code>ands xzr, x0, x1</code>	N/Z from the AND result; C/V cleared

Checking Flags After `cmp x0, x1`

- $x_0 == x_1$: $Z = 1$.
- $x_0 < x_1$ (signed): $N \neq V$.
- $x_0 < x_1$ (unsigned): $C = 0$.

Condition Codes: Signed vs. Unsigned Branches

Branch	Meaning	Flag Condition	Comparison Context
b.eq	Equal	$Z == 1$	Signed / Unsigned
b.ne	Not Equal	$Z == 0$	Signed / Unsigned
b.lt	Signed Less Than ($<$)	$N \neq V$	Signed
b.le	Signed Less or Equal ($\leq$)	$(Z == 1) \vee (N \neq V)$	Signed
b.gt	Signed Greater Than ($>$)	$(Z == 0) \wedge (N == V)$	Signed
b.ge	Signed Greater or Equal ($\geq$)	$N == V$	Signed
b.lo	Unsigned Lower ($<$)	$C == 0$	Unsigned
b.hs	Unsigned Higher or Same ($\geq$)	$C == 1$	Unsigned
b.hi	Unsigned Higher ($>$)	$(C == 1) \wedge (Z == 0)$	Unsigned
b.ls	Unsigned Lower or Equal ($\leq$)	$(C == 0) \vee (Z == 1)$	Unsigned

Usage Context

Unsigned conditions are appropriate when the program semantics treat the values as unsigned; they are also commonly used for address and range comparisons.

Control Flow: Branches & Selection

Compare-and-Branch Instructions: **cbz** and **cbnz**

- **cbz** / **cbnz** test a register directly and do not alter NZCV.
- They encode the zero test and branch in one architectural instruction instead of a separate compare and conditional branch.

Null Pointer Validation

```
cbz x0, .Lnull    // branch if null
ldr x1, [x0]      // dereference
```

Loop Counter Decrement

```
.Lloop:
sub x1, x1, #1    // count-
cbnz x1, .Lloop   // repeat if nonzero
```

Single-Bit Test: `tbz x0, #0, label` branches if bit 0 is zero.

Translating if-then-else Blocks

High-Level C Code

```
if (x > y) {  
    result = x - y;  
} else {  
    result = y - x;  
}
```

AArch64 Assembly Mapping

```
// Assume: x, y are signed int64_t  
// x0=x, x1=y, x2=result  
cmp x0, x1           // Compare x and y  
b.le .Lelse          // Invert test: jump if x <= y  
sub x2, x0, x1        // Then-body: result = x - y  
b .Lend              // Skip else block  
.Lelse:  
sub x2, x1, x0        // Else-body: result = y - x  
.Lend:
```

Assembly Pattern: Invert the high-level condition to fall through into the “then” path.

Loop Translation: Inverted Loop Pattern

High-Level C Loop

```
// Compute array sum
int64_t i = 0;
int64_t n = ...; // signed bound
while (i < n) {
    sum += arr[i];
    i++;
}
```

Bottom-Tested AArch64 Loop

```
// x0=arr, x1=n, x2=i, x3=sum
mov x2, xzr                // i = 0
mov x3, xzr                // sum = 0
b .Ltest                  // Jump to test first
.Lbody:
ldr x4, [x0, x2, lsl #3]   // Load arr[i]
add x3, x3, x4             // sum += arr[i]
add x2, x2, #1             // i++
.Ltest:
cmp x2, x1                 // i < n?
b.lt .Lbody               // Loop if true
```

Common Loop Layout: Placing the test at the bottom gives the steady-state loop body one loop-closing conditional branch.

Branch Misprediction & Conditional Selection (**cse1**)

- A mispredicted conditional branch discards younger speculative work; the cost is implementation-dependent.
- **cse1** chooses between two register operands from NZCV without changing control flow.
- **Trade-off:** Conditional selection avoids prediction for that decision but can extend a data-dependency chain.

Signed Maximum: $\text{result} = (x_1 > x_2) ? x_1 : x_2$

```
cmp x1, x2           // set NZCV from x1 - x2
cse1 x0, x1, x2, gt   // signed max; use hi for unsigned
```

Related Forms

`cset w0, eq` sets 0 or 1 from a condition; `cinc x0, x1, eq` conditionally increments x1.

Review and Self-Test Questions

Topic 4: Comprehensive Summary

- **Arithmetic & Shifts:** Regular register and immediate arithmetic includes shifted-register operands in many data-processing instructions.
- **Bitwise Operations:** `and`, `orr`, `eor`, `bic`, and bitfield instructions operate directly on packed binary data.
- **NZCV Flags:** Flag-setting arithmetic and aliases such as `cmp` and `tst` support signed and unsigned conditions.
- **Control Flow:** Compilers commonly arrange one path as fall-through and use bottom-tested loops where appropriate.
- **Conditional Selection:** `csel` and related forms can replace short conditional branches when data-dependent selection is preferable.

Self-Test Questions: Arithmetic & Logic

- ❶ **Shifted Addition:** What is the exact mathematical operation and resulting decimal value stored in `x0` after executing:

```
mov x1, #12  mov x2, #5  add x0, x1, x2, lsl #2
```

- ❷ **Bit Clear Operation:** If register `x1` contains `0xff` and `x2` contains `0x0f`, what value resides in `x0` after executing `bic x0, x1, x2`?
- ❸ **Signed vs. Unsigned Comparisons:** If `x0` contains `-1` (`0xffffffff_ffffffff`) and `x1` contains `1`, which branch will be taken after `cmp x0, x1: b.lt` or `b.lo`? Why?
- ❹ **Flag Updating:** Explain why `add x0, x1, x2` does not update the Zero (*Z*) flag even if the resulting sum is exactly zero.

Self-Test Questions: Control Flow & Selection

- ❶ **Bottom-Tested Loop Layout:** Why do compilers often place the conditional loop branch at the bottom of the loop body rather than at the top?
- ❷ **Branchless Translation:** Write an equivalent 3-instruction branchless AArch64 sequence using `cmp`, `neg`, and `csel` to implement:

```
int64_t abs_val = (val < 0) ? -val : val;
```

- ❸ **Compare-and-Branch Instructions:** When should an assembly programmer use `cbz x0, label` instead of `cmp x0, #0` followed by `b.eq label`?
- ❹ **Branch Misprediction Trade-off:** What factors determine whether `csel` or a conditional branch is faster?

Looking Ahead

Next Lecture Preview

Our Next Topic: Memory Access & AArch64 Addressing Modes

- **Load/Store Addressing:** Examining scaled and unscaled immediate offsets, register offsets, and effective-address calculation.
- **Indexed Addressing:** Using pre-indexed and post-indexed forms to combine memory access with base-register updates.
- **Data Width & Extension:** Working with byte, halfword, word, and doubleword transfers, including sign- and zero-extending loads.
- **Paired Transfers & Arrays:** Applying `ldp/stp`, pointer arithmetic, and addressing patterns used for arrays and structured data.